

Ayesha Azmi

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[Academic Website](#) | [Google Scholar](#) | [Orcid](#) | [linkedin](#) | [ResearchGate](#) | [GitHub](#)

RESEARCH INTEREST

AI-Assisted Software Engineering, Trustworthy and Secure Large Language Model (LLM) Systems, Software Security & Vulnerability Analysis, and LLM Security for Code Generation and Refactoring

EDUCATION

University of Asia Pacific

Bachelor of Science in Computer Science and Engineering
CGPA: 3.50/4.00 (WES: 3.55/4.00)

Dhaka, Bangladesh

May. 2016 – June. 2020

Thesis Title: HIDECHAIN: A User-Centric Secure Edge Computing Architecture for HealthCare IOT devices

RESEARCH EXPERIENCE

Research Intern

Dec. 2025 – Present

ELITE Research Lab LLC)

New York, USA

- Investigated and analyzed how AI-assisted code refactoring impacts software security, focusing on robustness and vulnerability risks.

Undergraduate Thesis Researcher

Dec. 2018 – Mar. 2020

University of Asia Pacific (Advisor: Prof. Abdullah Al Omar)

Dhaka, Bangladesh

- Designed and formulated HIDEchain, a user-centric secure edge computing architecture for healthcare IoT devices to address data security and privacy challenges.
- Analyzed and evaluated the proposed architecture for authorization, authentication, privacy, and data integrity using blockchain-based transaction auditing.

INDUSTRY EXPERIENCE

Software Engineer

Jul, 2023 – Present

Brain Station 23 PLC

Dhaka, Bangladesh

- Engineered scalable backend and distributed systems for telecom, fintech, and banking platforms using Java/Spring ecosystem.
- Developed fault-tolerant microservices, caching pipelines, and secure payment workflows serving large-scale production traffic.
- Engineered and optimized distributed systems, backend services, and mobile applications to improve reliability, performance, and security.

Software Engineer (Online)

Oct. 2020 – Jan. 2023

Lexoro GmbH)

Mannheim, Germany

- Developed scalable, fault-tolerant data systems—including an offline-first geospatial tracker and an automated medical data pipeline—to ensure reliability, consistency, and efficiency in large-scale environments.

SELECTED PROJECTS

Retail Distribution & Sales Platform (Telecom)

Oct, 2024 – Present

Tech: Java, Spring, jboss, Redis, MySQL, PostgreSQL, Oracle, Bash Script

Dhaka, Bangladesh

- Optimized distributed systems using Redis-based caching and pipelines, achieving 99.99% uptime and sub-second response for 100K+ concurrent users.
- Improved scalability and fault tolerance, reducing latency and increasing customer engagement by 15%.

Internet Banking

June, 2024 – Sept, 2024

Tech: Spring boot, Kafka, MySQL, Apache Velocity Engine

Dhaka, Bangladesh

- Architected secure financial systems integrating real-time core banking synchronization and automated data pipelines.

High-Availability Subscription Payment System (bKash)

July, 2023 – Feb, 2024

Tech: Spring Boot, PostgreSQL, Docker, Vue.js, Kafka

Dhaka, Bangladesh

- Engineered secure high-availability payment services supporting millions of users with adaptive rate limiting and real-time processing.

PUBLICATIONS & THESIS

Security Implications of LLM-Assisted Code Refactoring: A Mini Review

Manuscript in preparation

HIDEchain: A User-Centric Secure Edge Computing Architecture for Healthcare IoT Devices

Undergraduate Thesis, University of Asia Pacific, 2020

LANGUAGE PROFICIENCY

IELTS Academic: Overall Band Score: 7.0
Listening: 7.5
Reading: 7.5
Writing: 6.0
Speaking: 6.0

TECHNICAL SKILLS

Programming: Java, SQL, Bash, Python
Databases & Storage: PostgreSQL, MySQL, Oracle, Redis,
Framework & Libraries: Spring Boot, Hibernate, Spring Security, Kafka
Tools: Docker, Git, Azure, JIRA
Research Writing Tool: LaTeX

HONORS & AWARDS

Spring, 2018 – Dean’s Award

LEADERSHIP, MENTORING & TEACHING

Invited Speaker	oct, 2025
<i>TechSisters</i>	<i>Dhaka, Bangladesh</i>
• Delivered a session on professional development, technical versatility, and overcoming barriers for women in technology.	
Section Leader	Spring, 2025
<i>Stanford University’s Code in Place</i>	<i>California, USA</i>
• Mentored a cohort of 10+ global students in Stanford’s introductory Python programming course, leading weekly discussion,clarifying programming concepts, and assisting with assignments and problem-solving.	
Mentor	Spring, 2023
<i>Technovation</i>	<i>USA</i>
• Guided a team of students through the development of a mobile application, providing technical assistance and project management support.	

CERTIFICATIONS

2025 – Empowering Women in Tech (TechSisters Kickstart)
2023 – Sanford University’s Code in Place

REFERENCES

A S M Touhidul Hasan
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